

Bond-by-bond loss-absorption results

The programme produced 14 positive, 15 negative and 1 unresolved results for 30 bond series. These are qualified readings of dated offering documents. Preferred-share securities are excluded. A positive identifies principal write-down or compulsory common-share conversion, including disclosed statutory bail-in. A negative requires affirmative debt/repayment terms and no unresolved potentially qualifying candidate in the declared source set. Ordinary creditor-approved restructuring is treated separately.

1. Tesco treasury 3.500% senior notes due 2033

ISIN / identity: XS3201918409

Result: No — non loss absorption

No feature identified within the examined offering terms: cash principal repayment is provided ([tesco-2025-final](#), PDF pp.2-4); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

2. Veolia 3.690% senior notes due 2031

ISIN / identity: FR0014017P12

Result: No — non loss absorption

No feature identified within the examined offering terms: cash principal repayment is provided ([veolia-2026-base](#), PDF p.79); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

3. Veolia 4.122% senior notes due 2036

ISIN / identity: FR0014017P04

Result: No — non loss absorption

No feature identified within the examined offering terms: cash principal repayment is provided ([veolia-2026-base](#), PDF p.79); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

4. Deutsche Bank 6.75% perpetual AT1, December 2025

ISIN / identity: DE000A460DG7

Result: Yes — loss absorption

Yes: the principal can be reduced or cancelled under the contractual terms ([deutsche-at1-2025](#), PDF p.9). The conditions are retained in the cited evidence; the power need not have been exercised.

5. Sogécap 6.25% perpetual restricted tier 1, July 2025

ISIN / identity: FR00140112W8

Result: Yes — loss absorption

Yes: the principal can be reduced or cancelled under the contractual terms ([sogecap-rt1-2025](#), PDF p.72). The conditions are retained in the cited evidence; the power need not have been exercised.

6. UBS 5.600% SGD perpetual AT1 (June issue and July tap)

ISIN / identity: CH1357852636

Result: Yes — loss absorption

Yes: the bond can be compulsorily converted into common/ordinary shares under the contractual terms ([ubs-sgd-at1-2024-final-published](#), PDF p.28). The conditions are retained in the cited evidence; the power need not have been exercised.

7. Barclays 7.625% perpetual AT1, first reset 2035

ISIN / identity: US06738EDC66

Result: Yes — loss absorption

Yes: the bond can be compulsorily converted into common/ordinary shares under the contractual terms ([barclays-at1-2025](#), PDF p.57). The conditions are retained in the cited evidence; the power need not have been exercised.

8. HSBC 6.750% perpetual AT1, first reset 2031

ISIN / identity: hsbc-at1-2026-2031

Result: Yes — loss absorption

Yes: the bond can be compulsorily converted into common/ordinary shares under the contractual terms ([hsbc-at1-2026-specific](#), PDF p.1). The conditions are retained in the cited evidence; the power need not have been exercised.

9. HSBC 7.000% perpetual AT1, first reset 2036

ISIN / identity: hsbc-at1-2026-2036

Result: Yes — loss absorption

Yes: the bond can be compulsorily converted into common/ordinary shares under the contractual terms ([hsbc-at1-2026-specific](#), PDF p.1). The conditions are retained in the cited evidence; the power need not have been exercised.

10. Standard Chartered 7.625% perpetual AT1, January 2025

ISIN / identity: US853254DF47, USG84228GP72

Result: Yes — loss absorption

Yes: the principal can be reduced or cancelled under the contractual terms ([standard-chartered-at1-2025-executed-deed](#), PDF pp.42-43). The conditions are retained in the cited evidence; the power need not have been exercised.

11. Apple Inc. 4.000% Notes due 2028

ISIN / identity: apple-senior-2025.1

Result: No — non loss absorption

No feature identified within the examined offering terms: cash principal repayment is provided ([apple-senior-2025](#), PDF p.16); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

12. Apple Inc. 4.200% Notes due 2030

ISIN / identity: apple-senior-2025.2

Result: No — non loss absorption

No feature identified within the examined offering terms: cash principal repayment is provided ([apple-senior-2025](#), PDF p.16); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

13. Apple Inc. 4.500% Notes due 2032

ISIN / identity: apple-senior-2025.3

Result: No — non loss absorption

No feature identified within the examined offering terms: cash principal repayment is provided ([apple-senior-2025](#), PDF p.16); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

14. Apple Inc. 4.750% Notes due 2035

ISIN / identity: apple-senior-2025.4

Result: No — non loss absorption

No feature identified within the examined offering terms: cash principal repayment is provided ([apple-senior-2025](#), PDF p.16); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

15. Duke Energy Corporation 5.45% Senior Notes due 2034

ISIN / identity: duke-senior-2024.1

Result: No — non loss absorption

No feature identified within the examined offering terms: cash principal repayment is provided ([duke-senior-2024](#), PDF p.15); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

16. Duke Energy Corporation 5.80% Senior Notes due 2054

ISIN / identity: duke-senior-2024.2

Result: No — non loss absorption

No feature identified within the examined offering terms: cash principal repayment is provided ([duke-senior-2024](#), PDF p.15); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

17. The Southern Company Series 2025A 6.50% Junior Subordinated Notes due 2085

ISIN / identity: southern-junior-2025a.1

Result: No — non loss absorption

No feature identified within the examined offering terms: cash principal repayment is provided ([southern-junior-2025a](#), PDF p.10); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

18. Duke Energy Corporation 6.45% Fixed-to-Fixed Reset Rate Junior Subordinated Debentures due 2054

ISIN / identity: duke-junior-2024.1

Result: No — non loss absorption

No feature identified within the examined offering terms: cash principal repayment is provided ([duke-junior-2024](#), PDF p.12); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

19. NatWest Markets 4.789% senior notes due 2028

ISIN / identity: USG6382RGD47, US63906YAM03

Result: Yes — loss absorption

Yes: the principal can be reduced or cancelled under a disclosed statutory resolution power ([nwm-2025-base](#), PDF p.69). The conditions are retained in the cited evidence; the power need not have been exercised.

20. BPCE USD SOFR + 1.03% senior preferred notes due 2030

ISIN / identity: FR0014014YQ1

Result: Yes — loss absorption

Yes: the principal can be reduced or cancelled under a disclosed statutory resolution power ([bpce-2025-base](#), PDF p.150). The conditions are retained in the cited evidence; the power need not have been exercised.

21. CBA EUR ESTR + 0.26% senior notes due October 2026

ISIN / identity: XS3206613666

Result: No — non loss absorption

No feature identified within the examined offering terms: cash principal repayment is provided ([cba-2025-6700-final](#), PDF pp.2-4); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

22. Compass Group 3.500% senior notes due 15 January 2035

ISIN / identity: XS3232968985

Result: No — non loss absorption

No feature identified within the examined offering terms: cash principal repayment is provided ([compass-2035-final](#), PDF pp.2-3); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

23. ING Groep 7.000% perpetual AT1, first reset November 2032

ISIN / identity: US456837BT90

Result: Yes — loss absorption

Yes: the bond can be compulsorily converted into common/ordinary shares under the contractual terms ([ing-at1-us456837bt90](#), PDF p.36). The conditions are retained in the cited evidence; the power need not have been exercised.

24. Shell International Finance 1.000% senior notes due 10 December 2030

ISIN / identity: XS2228291279

Result: No — non loss absorption

No feature identified within the examined offering terms: cash principal repayment is provided ([shell-2030-final](#), PDF pp.2-3); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

25. Danske Bank DKK1.25bn perpetual AT1, Series 4, August 2026

ISIN / identity: DK0030577200

Result: Yes — loss absorption

Yes: the bond can be compulsorily converted into common/ordinary shares under the contractual terms ([danske-at1-2026-base](#), PDF p.110). The conditions are retained in the cited evidence; the power need not have been exercised.

26. Lloyds 6.625% perpetual AT1, first reset September 2035

ISIN / identity: US539439BF59

Result: Yes — loss absorption

Yes: the bond can be compulsorily converted into common/ordinary shares under the contractual terms ([lloyds-at1-2025-yahoo-filing](#), HTML text unit 1). The conditions are retained in the cited evidence; the power need not have been exercised.

27. Banco Santander 6.000% perpetual AT1, July 2025

ISIN / identity: XS3100756637

Result: Yes — loss absorption

Yes: the bond can be compulsorily converted into common/ordinary shares under the contractual terms ([santander-at1-2025](#), PDF p.103). The conditions are retained in the cited evidence; the power need not have been exercised.

28. Unilever Capital Corporation 2.750% notes due 22 May 2030

ISIN / identity: XS3081333547

Result: No — non loss absorption

No feature identified within the examined offering terms: cash principal repayment is provided ([unilever-2030-final](#), PDF pp.2-4); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

29. ABN AMRO 5.750% perpetual AT1, February 2025

ISIN / identity: XS3004202811

Result: Yes — loss absorption

Yes: the principal can be reduced or cancelled under the contractual terms ([abn-at1-2025](#), PDF p.1). The conditions are retained in the cited evidence; the power need not have been exercised.

30. Enel Finance International 2.625% senior notes due 24 February 2028

ISIN / identity: XS3008888953

Result: Unresolved

No supported yes/no answer: 1 potentially relevant clauses require scope/meaning resolution

What was checked

585 prospectus tests passed, including 443 tests for this classifier. The independent specification checked all 256 known/unknown/conflict input states; two SMT equivalence obligations passed and four deliberately wrong formal rules were detected. RuleIR/Java and Catala executed the same 30 bond input sets, plus 24 formal challenge cases in each of 2 runs, for 156 native executions. Kernel-checked lowering and source integrity establish conditional software properties, not legal interpretation.

Every one of the 7,584 emitted evidence records was rebound to its preserved source text. 20 faults involving actual PDFs/text were rejected or left unresolved. The reader repairs were checked against action, polarity, issue, definition, source and monetary-relation challenges. Previously inspected documents are development cases. The new frozen version processed these fresh cases: abn-amro-at1-2025 (yes), enel-senior-2028 (unresolved). No source-answer labels were used.

Qualifications and remaining work

The binary result is the existence of the requested prospectus feature. Positive rows can still contain unresolved subsidiary clauses: a sufficient witness establishes the feature without establishing every possible mechanism or its limits. The full JSON retains those open clauses. A negative is conditional on the declared document selection and bounded English analysis; it is not a proof that all incorporated contracts, later amendments and applicable law have been found. The system cannot independently prove unrestricted English meaning or correctness for every unknown future document. It makes that qualification explicitly and uses missing-evidence states rather than inventing certainty.

Standard Chartered uses an issuer-published executed trust deed because the retained circular carries subject-to-completion language. Deutsche Bank's German terms prevail; the English translation has not been independently proved equivalent. Lloyds uses a retained final filing mirror: official issuer-PDF acquisition failed, and byte identity with EDGAR is not established. Some US/HSBC series use filing/series identities because their ISINs were not independently resolved. Issuer names, domicile and rank are metadata, never classification rules.

Shell now has an exact repayment derivation: £1,000 final redemption per £1,000 calculation amount. The definitions and money fields, source positions, selected issue and generated explanation are checked together. A separate checker evaluates the decision and rejects inconsistent or altered evidence. It shares the bounded English constructions with the reader; it is independent of the decision implementation, not an independent proof of natural-language interpretation.

Next work: strengthen semantic subject/condition resolution, close incorporated-document and amendment dependencies, validate authoritative language alignment, and run further frozen document-family challenges. Treat future source or law changes as reasons to re-acquire and re-evaluate. A negative here does not mean low investment risk or transaction permission.

[Machine-readable results](#) · [CSV](#) · [verification](#) · [reviewed execution plan](#) · [source freeze](#)

What the repairs establish

The repaired reader distinguishes an obligation to report a possible conversion from an obligation to convert the bond. It also keeps negation attached to the action and rejects principal-loss evidence belonging to a different named series. These repairs remove the

recorded counterexamples. They do not prove that every unfamiliar sentence will be interpreted correctly.

A positive reason now comes from the same source witness as the decision. Standard Chartered's reason uses the conditional principal write-down in its executed deed, on PDF pages 42–43. The Newco reorganisation provision cannot supply that witness. Duke's 2054 repayment reason must refer to that series; the 2034 clause cannot establish its repayment terms. Cancellation after ordinary-share delivery and language about other securities reducing recovery are no longer treated as principal write-downs.

Shell's final terms specify a £1,000 calculation amount on PDF page 2 and £1,000 final redemption per calculation amount on page 3. The reader now compares those exact amounts and their currencies. That closes the known repayment-unit gap. Tesco's uncompleted forms on pages 36–58 of its base circular are searched as programme options, while the selected final terms supply the issue's completed terms. Both changes depend on explicit document structure and source fields, rather than an expected negative answer.

NatWest and BPCE remain positive because their terms disclose regulatory powers to reduce principal. BPCE's definition of Amounts Due is necessary: it expressly includes principal. A definition covering only interest would not establish the requested feature. CBA's negative continues to exclude ordinary creditor-approved restructuring from this feature test. That exclusion does not mean that principal can never be lost.

The report answers whether the selected offering documents identify principal write-down or compulsory conversion to common shares. It does not decide SFC product complexity, client suitability, sanctions compliance or transaction permission. A subordinated bond can have a qualified negative feature result while other selling restrictions still apply.

The independent checks establish the formal decision, backend agreement and integrity of source-to-explanation links. The English constructions are shared between extraction and witness checking, so that agreement cannot prove their meaning. All real-document answers remain qualified. Document completeness, authoritative-language equivalence, later amendments and performance on unknown future instruments remain unresolved. No human answer labels or legal accuracy score are used.

The earlier frozen challenge left Santander and Unilever unresolved. Both became development cases during the subsequent repair. Santander's operative Condition 5.1(c), on PDF page 103, requires the Bank to convert the securities into Common Shares on the Trigger Event without holder consent. The reader now keeps the numbered action with its governing actor and obligation. Common Shares are defined as ordinary shares on PDF page 86. A prohibition on optional holder conversion does not negate that mandatory mechanism. The resulting positive identifies this sufficient feature; subsidiary unresolved clauses remain in the JSON and are not thereby settled.

Unilever's 16 May 2025 memorandum was recovered from the public asset endpoint corresponding to the issuer link. Its content hash matches the filename in that link; the issuer endpoint itself continued to return HTTP 403. The selected 20 May 2025 final terms name that exact edition. Their EUR 1,000 calculation amount and EUR 1,000 final redemption per calculation amount supply the cash repayment relation. All 131 pages of the memorandum are examined, with its uncompleted programme forms distinguished from the selected issue terms. No qualifying feature was identified within those retained disclosures. The separate trust deed, incorporated financial and constitutional documents, later amendments and complete applicable law have not been reconstructed. The negative therefore remains a bounded source reading. Neither repaired case counts as evidence from a previously unseen issuer family.

The next frozen challenge used ABN AMRO's AT1 circular and Enel's senior-note final terms, base prospectus and supplement. ABN AMRO supplies a principal-loss witness. Enel remains unresolved: its final terms put a parenthetical instruction between "Calculation Amount" and EUR 1,000, without the colon required by the frozen reader. The final redemption field refers to the calculation amount, but the reader cannot establish that unit. The document's senior ranking cannot supply the missing relation. Vodafone preceded Enel in the recorded acquisition order; failures to retrieve its final terms and base prospectus remain preserved. These two new families expose the method's coverage directly: one supplies a supported feature witness and one requires a further parsing repair. They do not supply a population accuracy estimate.